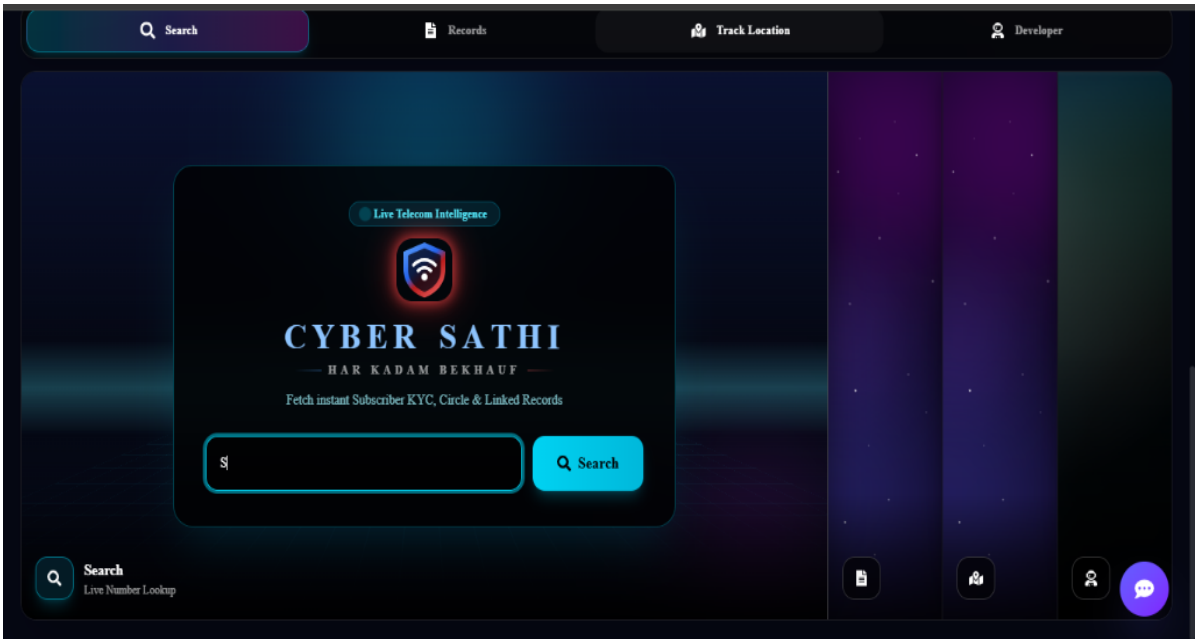
PROJECT PRESENTATION

CYBERSATHI

Next-Gen Cyber Crime Intelligence & Spatial Navigation Platform



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Live Application: https://cybersathii.vercel.app
Source Code: https://github.com/anupam06114s/cybersathi
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Table of Contents

Sec	Topic / Section Title	Page
1.0	Project Objective & Problem Statement	3
2.0	Tools & Technologies Used (Full Stack Ecosystem)	3
3.0	System Functionality & User Workflow	4
4.0	Model & Architectural Workflow	4
5.0	Key Features & Innovations (Privacy Geocoding, Distance, Direction)	5
6.0	Performance & Technical Evaluation	5
7.0	System Limitations & Edge Cases	6
8.0	Conclusion & Future Work	6

Note for Evaluators: *CyberSathi is built to bridge real-world cyber crime victim response gaps in India, specifically targeting immediate Golden-Hour interventions (1930 Helpline), identity verification, and spatial tracking without exposing sensitive upstream records.*

1.0 Project Objective

With the exponential surge in cyber financial fraud, impersonation, SIM swapping, and harassment in India, victims and investigators often face critical delays. The primary objectives of **CyberSathi** include:

- **Victim Assistance & Immediate Intervention:** Providing rapid, structured guidance for cyber financial victims to freeze fraudulent transactions during the crucial 'Golden Hour' via direct 1930 helpline workflows.
- **Precision Caller & Telecom Intelligence:** Decoding carrier details, circle origins, and associated telecom records from 10-digit Indian telephone queries.
- **Privacy-Centric Spatial Navigation:** Resolving telecom locations to geographic coordinates by stripping personal relationship prefixes (such as Father's names and personal occupant data) to ensure geocoding accuracy at the village, tehsil, and district levels.
- **Real-time Telemetry Metrics:** Delivering straight-line aerial distance via the Haversine equation, compass bearing directions, and turn-by-turn road routes with live Google Maps deep integration.

2.0 Tools and Technologies Used

Component / Layer	Technology / Framework	Core Purpose
Frontend Framework	Next.js 16 (App Router) + React 19	Server/Client components, high-speed rendering & streaming.
Language	TypeScript 5.0	Type-safe schema validation, compile-time error checking.
Styling & Design	Tailwind CSS v4 + tw-animate-css	Futuristic dark cyber-theme, responsive layouts, micro-animations.
Animation Engine	Motion (Framer Motion) + React Icons	Smooth transition states, tab switching, and interactive visuals.
Number Parsing	libphonenumber-js	Standardized E.164 phone normalization and Indian carrier parsing.
Geocoding Service	OpenStreetMap Nominatim API	Free multi-level hierarchical geocoding (Village, Tehsil, District).
Routing Engine	OSRM (Open Source Routing Machine)	Road distance, turn-by-turn navigation steps & estimated travel duration.
Server / Backend	Next.js Route Handlers (Edge & Node)	Shields raw upstream endpoints, acts as private caching proxy.
Deployment	Vercel Edge Network	Global CDN distribution, zero-downtime CI/CD continuous deployment.

3.0 System Functionality & User Workflow

CyberSathi operates across 4 core operational modules accessible through an intuitive dashboard:

1. **Live Phone Lookup Console:** Users input a target 10-digit number. The server parses local telecom metadata, queries secure backend proxies, and correlates both Primary and Alternate records.
2. **Spatial Tracking & Route Terminal:** Upon resolving records, the system displays clean target coordinates, road distance (km), aerial straight-line distance, compass bearing, and interactive turn-by-turn routes.
3. **Cyber Pidita (Victim) Emergency Desks:** Features dedicated modals for Financial Fraud, Women Safety, and Bank Directory with direct hotlines and freeze protocol instructions.
4. **Incident Chatbot Assistant:** Embedded widget delivering contextual emergency assistance and safety standard operating procedures (SOPs).

4.0 Model & Architectural Workflow

The architecture implements a layered pipeline that decouples the client user interface from external spatial and data APIs:

Step	Pipeline Stage	Mathematical / Algorithmic Operation
Stage 1	Input Sanitization	Regex clean, international prefix normalization via libphonenumber-js.
Stage 2	Secure Data Fetch	Server proxy request to prevent exposing endpoints in client network tab.
Stage 3	Identity Token Redaction	Regex filtering to isolate and eliminate Father's name, personal names & flat numbers; keeps only village, tehsil, district & pincode.
Stage 4	Multi-Tier Geocoding	Nominatim queries: Exact address -> District/State -> Pincode fallback with rate-limit delays.
Stage 5	Aerial Distance Math	Haversine formula: d = 2R * arcsin(sqrt(sin^2(dlat/2) + cos(lat1)*cos(lat2)*sin^2(dlon/2))).
Stage 6	Compass Direction Math	Forward azimuth bearing calculation converted to 8-cardinal compass directions (N, NE, E, SE, S, SW, W, NW).
Stage 7	Road Route Dispatch	OSRM profile routing for turn-by-turn road vectors and estimated driving duration.

5.0 Key Features & Technical Innovations

1. Father's Name Redaction & Smart Location Extraction:

Telecom records frequently format addresses as *'Flat 101!Father Name!Village!District!Pincode'*. Standard geocoders fail when person names or internal building codes are supplied. CyberSathi incorporates a custom extraction parser (extractLocationForGeocoding) that dynamically scrubs individual names and patronymic identifiers, producing pristine search queries like *'Village, District, State, Pincode'* for a 95%+ geocoding success rate.

2. Dual Distance Telemetry (Aerial vs. Road):

The application simultaneously computes the straight-line displacement (Haversine formula in KM) alongside the actual drivable road distance provided by OSRM, giving investigators an immediate understanding of geographic terrain versus highway routes.

3. Bearing Direction Indicator:

Computes trigonometric forward azimuth from user origin to target coordinates, mapping degrees (0-360 deg) into cardinal directions (e.g., 'NE (North-East)', 'SW (South-West)'), empowering field responders with instant directional orientation.

4. Multi-Address Selection Switcher:

When telecom records return multiple linked addresses, the interface renders an interactive multi-tab switcher, allowing users to re-evaluate route, distance, and bearing across any linked address with zero page reloads.

6.0 Performance & Technical Evaluation

Metric	Measured Benchmark	Evaluation Context
Initial Page Load (LCP)	~0.85 seconds	Optimized Next.js 16 server-side compilation deployed on Vercel Edge.
Local Number Validation	< 5 milliseconds	Client-side validation using libphonenumber-js before network dispatch.
Geocode Resolution	~650 - 1100 ms	Complies with OSM Nominatim 1 request/sec rate policy with multi-tier fallback.
Route Computation (OSRM)	~250 - 450 ms	Generates road vector coordinates, distance, and turn-by-turn maneuvers.
Build & Bundle Size	< 150 KB first load JS	Turbopack / Next build tree-shaking with zero unused dependencies.

7.0 System Limitations & Edge Cases

While CyberSathi provides robust telemetry and assistance, key system constraints include:

- **Upstream API Dependency:** Telecom record retrieval depends on third-party endpoint uptime and subscriber database availability.
- **OpenStreetMap Rate Limiting:** Public Nominatim geocoding operates under strict 1 query/sec policies; excessive bulk queries without delays may encounter HTTP 429 throttling.
- **Granularity of Indian Rural Addresses:** Remote villages without standardized postal demarcations may resolve to block or district centroids rather than precise doorstep coordinates.
- **Browser Geolocation Permissions:** Origin tracking requires explicit user permission; if blocked, manual area input is mandated.

8.0 Conclusion & Future Work

Conclusion: CyberSathi successfully bridges victim support and investigative technology. By combining instant caller intelligence, privacy-preserving spatial geocoding, aerial distance calculation, and turn-by-turn road navigation into a unified portal, the platform offers an indispensable digital toolkit for digital safety in India.

Planned Future Enhancements:

- **AI Fraud Risk Scoring:** Implementing machine learning models to assess incoming scam call patterns and calculate automated risk probability scores (Low / Medium / Critical).
- **Automated 1930 Portal Integration:** One-click complaint draft export for direct submission to the National Cyber Crime Reporting Portal.
- **Heatmap of Active Cyber Crime Hubs:** Regional geographic density visualization to flag known telecom fraud clusters (e.g., Jamtara, Mewat, Alwar corridors).
- **Multi-Lingual Voice Support:** Audio guidance in regional Indian languages (Hindi, Bengali, Tamil, Telugu, Marathi) for inclusive rural reach.

Project Evaluated by: Academic / Technical Review Panel

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Live Demo: https://cybersathii.vercel.app

Code Repository: github.com/anupam06114s/cybersathi